



Ninth National Aerospace Conceptual Design Competition

(NACDeC-IX)

Initial Design Report

Submitted by

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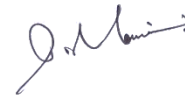


31 March 2026

Certificate

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Abstract

This Initial Design Report outlines the conceptual design and sizing of a Tactical Hunting and Reconnaissance (THAR) Unmanned Aerial Vehicle (UAV) designed to perform continuous border surveillance along the Indian border in the Thar Desert. The UAV is expected to achieve an operational range of approximately 1200 km and an endurance of approximately 11 to 13 hours while carrying a payload capacity of 15 Kg at an operating altitude of 3000 meters. The platform is designed for operations in constrained environments, incorporating Vertical Take Off and Landing (VTOL) capability to eliminate the need for conventional runway infrastructure.

Aerodynamic design of wing is based on the NACA 23012 airfoil, selected for its favourable lift characteristics and stability. A high-wing, tapered planform is adopted to enhance lift-to-drag ratio and reduce induced drag, improving cruise efficiency. An inverted U-boom tail configuration is selected to ensure enhanced stability, effective control, and structural efficiency. The initial performance analysis of VTOL configuration thrust required is 2207.5 N and at a cruise velocity of 40 m/s indicates a thrust requirement of 200 N and a power requirement of 8 kW.

The proposed configuration features a hybrid propulsion system consisting of a quad-motor electric VTOL setup for vertical lift and transition, combined with a pusher propulsion system using a tri-blade propeller for efficient forward cruise. The results demonstrate a feasible and efficient UAV configuration that successfully balances VTOL capability, aerodynamic performance, and long-endurance mission requirements, providing a strong foundation for further detailed design and development.

Keywords: Tactical Hunting, Reconnaissance, Unmanned Aerial Vehicle (UAV), Border Surveillance, Patrol Missions, Neutralisation payload

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Nomenclature

Acronyms

AR	Aspect Ratio
CG	Center of Gravity
EO/IR	Electro-Optical/Infrared
GPS	Global Positioning System
GNSS	Global Navigation Satellite System
ISR	Intelligence, Surveillance and Reconnaissance
MAC	Mean Aerodynamic Chord
MTOW	Maximum Take-Off Weight
RPM	Revolutions per minute
UAS	Unmanned Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing

Symbols

b	Wingspan (m)
$\bar{c}$	Mean Aerodynamic Chord (m)
C_D	Drag Coefficient
C_L	Lift Coefficient
C_{Lmax}	Maximum Lift Coefficient
D	Drag Force (N)
E	Endurance (sec)
J	Advance Ratio
l_H	Horizontal Tail Moment Arm (m)
l_V	Vertical Tail Moment Arm (m)
P	Power (W)

S	Wing Area (m^2)
S_H	Horizontal Tail Area (m^2)
S_V	Vertical Tail Area (m^2)
T	Thrust (N)
V	Velocity (m/s)
V_H	Horizontal Tail Volume Coefficient
V_V	Vertical Tail Volume Coefficient
W	Weight (N)
W_i	Initial Weight (N)
W_f	Final Weight (N)
W/S	Wing Loading (N/m^2)

Greek Symbols

α	Angle of Attack (deg)
ρ	Air Density (Kg/m^3)
η_p	Propeller Efficiency
λ	Taper Ratio

Subscripts

L	Lift
D	Drag
max	Maximum
p	Propeller
r	Root
H	Horizontal tail
V	Vertical tail
i	Initial
f	Final

Chapter 1 Introduction

Unmanned Aerial Vehicles (UAVs) have become a significant part of the modern defence mechanism because of their potential to carry out long-duration operations without putting human lives at risk. The rising need for uninterrupted Intelligence, Surveillance, and Reconnaissance (ISR) operations, especially in difficult terrains such as the desert, has led to the development of advanced UAVs that provide flexibility along with aerodynamic efficiency.

The report discusses the conceptual design and analysis of a Tactical Hunting and Reconnaissance UAV named THAR UAV, which is designed for the NACDeC-IX competition. The objective of this UAV is to carry out continuous border surveillance in the Thar Desert, detect suspicious activities using EO/IR sensors, and also deploy a non-lethal neutraliser. This UAV has to take off and land without any conventional runway within a confined space. Therefore, VTOL is a crucial design consideration.

In order to satisfy these requirements, a hybrid configuration for the UAV is adopted, in which distributed electric propulsion is used for VTOL flight, while a fuel-powered pusher propulsion configuration is used for efficient cruise flight. The adopted configuration consists of a high-wing configuration, twin boom tail configuration, and a streamlined fuselage configuration to ensure aerodynamic efficiency, structural simplicity, and payload accommodation. The design process adopted is a mission-oriented approach in which performance, stability, propulsion, and structural requirements are considered simultaneously to ensure a balanced design.

The report will also cover a comprehensive analysis of the design drivers, configuration selection, aerodynamic characteristics, propulsion system sizing, and preliminary stability considerations. Initial sizing will be performed using established empirical methods and constraint-based analysis to ensure the UAV satisfies the required criteria.

The proposed THAR UAV design is intended to provide a reliable, efficient, and adaptable solution for long-endurance surveillance operations in harsh environments, yet simple, transportable, and operationally effective. It is worth noting that all parameters and results provided in this report relate to the conceptual design phase and are based on preliminary calculations that will be adjusted in subsequent design stages.

Chapter 2 Literature review

2.1.Literature Survey of Similar UAVs

Unmanned Aerial Vehicles (UAVs) have become more important in modern tactical defence operations due to their ability to carry out persistent surveillance, intelligence and reconnaissance missions where real-time awareness and rapid deployment are crucial while minimizing risks to human personnel. The development and evolution of UAVs show a gradual shift from basic remotely operated UAV to highly autonomous and mission-optimized aerial vehicles that form an integral part of modern defence strategies. Initially designed for target practice and basic reconnaissance missions during early 20th-century wars, UAVs were basic and largely dependent on remote operation and minimal intelligence. During World War II, basic drones or remotely operated UAV, such as ‘radio-controlled UAV,’ were first introduced for training and experimental combat missions, thus laying the foundation for UAVs. These early systems, however, were limited by a lack of onboard intelligence, short endurance, and dependence on manual control [1].

The late 20th century was a significant period in the evolution of UAVs due to the addition of GPS navigation, real-time data communication, and better propulsion. This technological advancement led to the development of highly advanced MALE UAVs like the IAI Heron, which incorporated its capabilities of long endurance with advanced real-time surveillance and communication capabilities. The RQ-1 Predator and similar platforms have significantly changed the face of modern warfare by providing real-time video transmission and long-endurance ISR capabilities. This period has also seen the shift from dedicated reconnaissance UAVs to multi-role platforms that incorporate strike and sensor capabilities. This period marked the transition from simple reconnaissance drones to multi-role UAV systems capable of both surveillance and combat support [2]. As we enter the 21st century, UAV technology is advancing to highly autonomous and mission-specific designs. UAVs like MQ-9 Reaper showcase UAV capabilities such as long-endurance flights of more than 24 hours, high-altitude flights, and effective incorporation of ISR payloads to enable persistent surveillance of vast geographical areas. These platforms highlight the importance of aerodynamic efficiency, lightweight structures, and fuel-efficient propulsion in achieving persistent surveillance capability [1], [2]. At the same time, there is considerable progress in mini-UAVs/tactical drones, opening up UAV applications in border control.



Figure 2.1 : IAI Heron ^[3]

One of the major developments in the evolution of UAVs is the introduction of the hybrid VTOL (Vertical Take-Off and Landing) fixed-wing configuration, which overcomes the limitations of the rotary and fixed-wing UAV. The tactical reconnaissance role is a major application of the hybrid configuration, which can undertake efficient flight operations in a restricted environment. UAVs such as Quantum Systems Vector and Penguin C VTOL are examples of the latest developments showing how distributed electric propulsion and fuel-efficient cruise systems can be combined to achieve operational flexibility and extended endurance. More recently, UAV such as the JOUAV CW-80E UAV have shown the progression of hybrid VTOL UAV, allowing for both vertical take-off and efficient fixed-wing cruise. One notable achievement in this category is the ROC VTOL UAS, which showcases the scalability of the hybrid UAS concept, with the vehicle being able to achieve an endurance time of over 20 hours, along with significant payloads.



Figure 2.2 : JOUAV CW-80E ^[4]

Despite the considerable improvements made in the design and development of the UAV, the currently available designs show considerable trade-off in terms of endurance, range, and operational flexibility. Although the conventional fixed-wing UAVs show considerable performance in terms of endurance and range, they are limited by the requirement for take-off

infrastructure, whereas the VTOL UAVs show considerable operational flexibility with the compromise of aerodynamic efficiency. The development of the hybrid quad-motor VTOL-assisted fixed-wing UAV has partially overcome the limitations of the conventional fixed-wing UAVs, but most of the currently available designs are either limited in terms of range or do not show considerable optimization in terms of endurance for the accomplishment of the tactical reconnaissance missions. Considering the aforementioned limitations, the proposed Tactical Hunting and Reconnaissance (THAR) UAV has been designed with the quad-motor VTOL-assisted fixed-wing configuration with the pusher propulsion system, along with the integration of the twin-boom inverted U-tail arrangement and the tapered trailing edge-aligned wing, showing considerable aerodynamic stability and efficiency, with the ability to perform the tactical reconnaissance missions with the endurance of around 11 hours and the range of around 1200 km.

Table 2.1 : Comparison of Benchmark UAV for THAR UAV Design

Parameter	IAI Heron (STOL)	JOUAV CW-80E UAV (VTOL)
Payload (Kg)	250	25
Maximum Take off Weight (Kg)	1150	100
Cruise Speed (m/s)	≈ 45	≈ 28
Maximum Speed (m/s)	≈ 58	≈ 36
Service Ceiling (m)	10,000	6,000
Endurance (hr)	52	9
Range (km)	≈ 1000	≈ 110
Wing Span (m)	≈ 16.6	≈ 5
Take off Distance (m)	≈ 1000	0
Landing Distance (m)	≈ 1000	0

While the IAI Heron is a Conventional runway Based UAV, it has been considered for this study as a reference system based on its exceptional endurance and payload capabilities. this would be enabling a comparative evaluation of conventional fixed wing UAV s and new VTOL concepts in terms of operational flexibility and performance.

Chapter 3 Design Drivers

The Tactical Hunting and Reconnaissance (THAR) Unmanned Aerial Vehicle design requirements are based on the mission requirements and operational limitations. These drivers are the foundation of the conceptual design and are major factors in decisions that affect aerodynamic configuration, choice of propulsion system, structural design, payload integration and mission performance.

3.1.Mission Design Drivers

The main aim of the THAR UAV is to conduct sustained surveillance of the Thar Desert along the international border working out of distant positions. The UAV should be in a position to monitor the human activity both in the daytime and at night with Electro-Optical/Infrared (EO/IR) turret and transmitting images of the surveillance to a ground control station. Once suspicious activity is confirmed the UAV needs to release a neutraliser capsule that is non-lethal to temporarily incapacitate the target until the time the ground forces come up.

3.2.Operational Design Drivers

The UAV needs to work on sandy desert, and thus it is expected to be launched and recovered within a pre-launching space of 50 m x 50 m and this is without catapults and other heavy launch mechanisms. One of the primary operational drivers behind the design is independence from runway requirements. This is a significant constraint that has a major effect on the choice of a vertical or short take-off capable design and lightweight and modular airframe design.

3.3.Transportability

The whole UAV system should be portable so that it can be stacked into a 2 m x 3 m x 3 m truck container. Thus, the UAV should have a modular design enabling easy assembling and disassembling the UAV with structural integrity and aerodynamic performance.

3.4.Performance Design Drivers

The mission performance requirements are also important in the design process. The UAV should also be capable of attaining a total flight time that exceeds 9 hours with a cruise altitude of 3000 m above mean sea level. Moreover, the UAV should have adequate endurance to cover a large area of the international boundary in the Thar area and then go back to the point of deployment. To meet these needs, the UAV design should focus on aerodynamic efficiency, optimum propulsion performance as well as adequate on-board energy or fuel.

3.5. Payload Design Drivers

The UAV should include an EO/IR turret with a mass of around 5Kg in order to offer stabilized day and night capability with 360 azimuth field of view and downwards gaze. Also, the UAV will be required to carry four 1 Kg neutraliser capsules fitted onto a launcher system weighing around 6 Kg. The total mass of the payload now is about 15 Kg, which has a direct impact on the estimate of the UAV weight, the location of the center of gravity, and the structural design.

3.6. Environmental Design Drivers

There are other design challenges associated with environmental conditions in the Thar Desert. Under dusty atmospheric conditions and moderate humidity, the UAV is required to run with the temperature ranging between -10° C and 45° C. These environment factors also affect the choice of materials, cooling needs of avionics and propulsion systems as well as the shielding of delicate electronic parts.

3.7. Geometric and Structural Load Constraints

The UAV cannot weigh over 175 Kg at take-off and the wingspan should not be higher than 6 m. Besides, the UAV should be able to carry a maneuvering load factor of 1.3 g during flight missions. These limitations seriously affect wing size, structural integrity and aerodynamic design.

3.8. Avionics and Communication System

The avionics design of the UAV should be able to accommodate autonomous functionality and extended communication. The UAV should also have the necessary on-board systems that include GPS/GNSS-INO sensor of navigation, a flight control computer, and engine control unit and command and control real-time image transmission sensor of the EO/IR turret. The UAV is not under the visibility of the land; thus, a reliable data link system needs to be incorporated to provide constant communication between the UAV and the ground station.

The main design criteria of the THAR UAV are runway independent operation, ability to fly long distances, the integration of a payload that carries a surveillance and neutralisation capability, transportability in a standard truck container, and the ability to operate in the harsh desert conditions. These drivers underlie the other phases of conceptual design, configuration choice, aerodynamic analysis and propulsion system sizing.

Chapter 4 Design Philosophy

4.1. Overall Design Approach

The design of the THAR UAV is based on a mission-oriented conceptual design methodology, in which a design configuration is developed to meet a given requirement, rather than maximizing a given design parameter. The conceptual design of an UAV is a process in which a feasible design configuration is developed by integrating layout decisions with analytical estimations, such as weight, performance, and aerodynamic characteristics. UAV design is a compromise-oriented, iterative process in which a number of conflicting factors, including aerodynamics, propulsion, and structures, are traded off simultaneously [2].

This is an iterative process in the sense that there is a constant refinement of the design variables. This is because any changes in the various systems tend to have an impact on the performance of other systems as well. A systems approach is necessary in the sense that aerodynamics, propulsion systems, structure, and control systems are integrated in a unified manner. It is important to note that the aim of the conceptual design process is not to optimize any of the parameters such as speed and lift but rather to come up with a balanced solution that meets all the requirements effectively. As such, the philosophy behind the design is multidisciplinary optimization such that the end product is a feasible solution [5].

4.2. Mission-Oriented Design Philosophy

The UAV configuration is mainly influenced by the mission requirement, as the UAVs intended role will determine the aerodynamics, propulsion, and structural arrangements. In the case of long-endurance surveillance, the UAVs endurance is essential compared to speed and maneuverability, as the UAV is expected to provide continuous surveillance rather than making high-speed turns. Therefore, the UAV will need to cruise at moderate speeds with minimum power consumption and aerodynamic drag [2], [6].

The payload integration is an essential factor in UAV design, especially for EOIR sensors, which need an unobstructed field of view and suitable conditions for operation. The UAV has to be able to operate as a stable platform for sensing, with minimal disturbance. The payload positioning is an important factor in UAV configuration, especially in regard to center of gravity. Moreover, the requirement for runway-independent operation raises the requirement for VTOL, which affects the propulsion system as well as the overall design of the UAV. Other environmental factors, such as the desert environment, affect the design as well, requiring a

reliable system. Therefore, the design philosophy ensures that all the systems are integrated in line with the mission requirement, making the UAV both aerodynamically efficient and operationally effective [5], [6].

4.3.Configuration Selection Philosophy

The selection of UAV configuration is an essential part of the conceptual design phase of UAV development. In this case, the selection of the high-wing UAV is made in order to improve the lateral stability of the UAV using the pendulum effect. This is because the center of gravity is below the UAV wing [5]. In addition, the high-wing UAV is selected in order to provide increased ground clearance. This is essential for UAV operations in uneven terrain. The twin boom design allows the efficient integration of a propulsion system located at the rear, in addition to a direct load path from the tail to the wing, resulting in a reduced structural weight. The reason for the selection of the H-tail configuration is the well-understood aerodynamic characteristics and the ability to analyze stability and control analysis [5]. The central pod fuselage facilitates the accommodation of payload, avionics, and energy systems in an efficient manner, keeping in view the mass distribution and control of the center of gravity. The placement of major components, including batteries, propulsion, and payload, is designed in a manner that ensures that the center of gravity is within acceptable limits for all phases of flight, including VTOL, transition, and cruise flight. Thus, it is said that the final configuration is a well-balanced combination of aerodynamic, and functional factors [2].

4.4.Aerodynamic Efficiency Philosophy

The aerodynamic design philosophy is based on the maximization of lift-to-drag ratio for long endurance performance, as it is highly dependent on aerodynamic efficiency. Drag minimization and lift optimization are critical to reduce power requirements for cruise flight [5]. Increasing aspect ratio reduces induced drag and increases aerodynamic efficiency, but it also increases structural weight and bending stress, which is a compromise between performance and structural feasibility [6]. The tapered wing configuration is chosen to closely resemble an elliptical distribution of lift, minimizing induced drag. At the same time, it remains manufacturable and has a simple structure. The aerodynamic design includes features such as washout, enhancing stalling behaviour so that it begins at the wing's root. Aerodynamic design must also take into account the interaction with propulsion and structural systems, ensuring that improvements in efficiency do not adversely affect UAV feasibility. In this regard,

aerodynamic philosophy is based on balancing performance optimization with structural and operational constraints [2].

4.5.Propulsion and Energy Philosophy

The type of propulsion system is selected based on the mission requirements. This is because various phases of flight impose various performance requirements. The propulsion system of an UAV is designed to produce sufficient power for take-off, climb, and cruise. These phases of flight have different performance requirements [2]. The hybrid propulsion system integrates electric propulsion for VTOL and conventional propulsion for cruise flight. Electric propulsion is used for high thrust at low speeds, thus giving vertical take-off and landing capability, whereas conventional propulsion is used for high efficiency in terms of energy for long-duration cruise flights [6]. The strategy for managing this is through phase-wise optimization, where high power is used for short-duration VTOL flights and cruise power is used for long-duration flights. During this transition period of flight, both propulsion systems are used in tandem, thus facilitating a smooth transition from rotor lift to wing lift, thereby preventing aerodynamic instability in flight. For stability in flight, it is important to ensure that there is a smooth transition in both thrust and lift, thus preventing aerodynamic instability in flight. However, it is also important to note that there is an added weight factor in the case of hybrid propulsion systems, especially considering the battery and multiple propulsion units. The trade-off analysis is necessary in such cases. The propulsion philosophy is based on balancing operational flexibility and system efficiency [2].

4.6.Structural and Integration Philosophy

The structural design is focused on efficient loading distribution, lightweight properties, and compatibility with other parts of the UAV. In structural layout, consideration is given to the loading distribution so that aerodynamic and inertial loads can be efficiently transmitted through the airframe. In this UAV, the twin boom design offers a direct load path between the tail and the wing [2]. The fuselage structure provides for the integration of the payload, avionics, and power systems in a compact form, ensuring the distribution of mass and control of the center of gravity. In addition, the structure should provide for the distribution of propulsion loads, especially in VTOL modes [5], [6]. The manufacturability aspect is considered by adopting simple shapes and modular components, which simplify the manufacturing process and enhance the reliability of the structure. The modularity also provides ease of transportation, assembly, and maintenance in the field. The choice of materials

provides for the exposure to environmental factors such as high temperatures and exposure to dust, enhancing the durability and reliability of the structure. As a result, the structural concept provides a balance between strength, weight, and practicality [2], [6].

4.7.Operational and System Philosophy

The design is focused on simplicity, reliability, and maintainability, especially for long-endurance flights in hostile environments. Designs are made to reduce mechanical complexity and minimize failure probability for better performance [2]. The operational aspects of the system are also considered in the design. In this design, the landing system is designed as a support structure rather than a traditional landing gear arrangement because of VTOL operation [6]. The environmental factors such as dust and temperature are also considered in the design of the system for better performance. Stable performance is critical for unmanned UAV because of its aerodynamic properties [5].

4.8.Design Trade-Off Philosophy

The design process of the UAV is considered a compromise in various parameters such as performance, weight, efficiency, etc. In this context, it has been observed that improvement in one parameter leads to compromise in another parameter. For instance, the addition of VTOL capability adds weight and complexity to the system but removes the requirement for runway infrastructure [2]. In addition, the improvement in aerodynamic efficiency is compromised by the structural considerations. The final design is obtained by iterative refinement, satisfying all the design constraints. Hence, the design is considered an optimal compromise rather than the maximization of the parameters [6].

Chapter 5 Baseline Configuration

5.1.Wing Configuration

A high wing tapered configuration is selected for the UAV. A moderate aspect ratio of about 6 is selected for the UAV. It has a significant impact on the aerodynamics of the UAV. An increased aspect ratio results in an increased lift to drag ratio and reduced induced drag and hence increases the endurance of the UAV. However, the aspect ratio also has the disadvantage of increased structural weight and bending stress. Hence, the moderate aspect ratio is selected for the UAV in order to achieve a balance between aerodynamics and structures [2], [5]. The tapered wing planform is selected in order to improve the spanwise lift distribution of the UAV. An elliptical lift distribution provides the best induced drag but is difficult to manufacture [5]. The high-wing configuration is chosen to enhance lateral stability. Since the center of gravity is located below the wing, a pendulum effect is generated, which increases roll stability. This decreases the necessity for extra dihedral angle. In addition, high-wing configurations are used in UAVs to enhance ground clearance for easier flying [2]. In order to enhance stall characteristics, washout is used at the wing tips. This decreases the angle of attack at the tips and ensures that wing root stalls before the tip. Consequently, aileron effectiveness is preserved even when stall occurs. This is beneficial when flying at low-speed because aileron control is maintained. This is a common technique used in aerodynamic design. In conclusion, the selected wing configuration strikes a balance between aerodynamic performance, structural strength, and stability [2], [5].

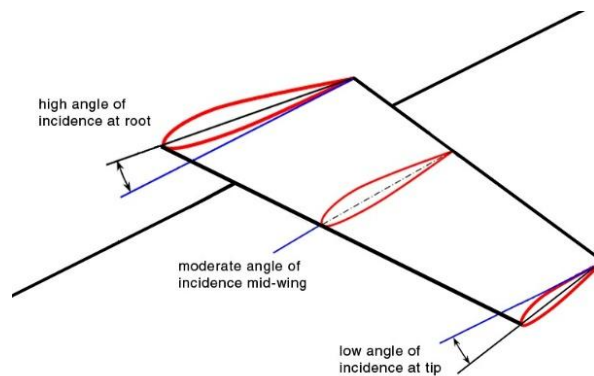


Figure 5.1 : Wing Selection [Courtesy: Wikipedia Washout(aeronautics)]

5.2.Tail Configuration

A twin boom tail configuration with a conventional H-tail configuration has been chosen for the THAR UAV to provide structural efficiency, aerodynamic stability, and integration with the

propulsion system. This twin boom configuration allows for the fitting of a rear-mounted pusher-type propeller without interference with the empennage standard UAV configuration design, and at the same time provides a direct and structurally efficient load path between the tail surfaces and the main wing. This also obviates the need for a long central fuselage, thereby facilitating transportability and modular assembly [2]. The conventional configuration for an H-tail, where there is a separate horizontal stabilizer and vertical stabilizer, is selected for its predictable and well-understood stability characteristics. The horizontal stabilizer is used for pitch and longitudinal stability, and the vertical stabilizer is used for directional stability, especially in crosswinds. The conventional configuration is advantageous over other configurations such as the V-tail configuration or inverted V-tail configuration in that it has decoupled pitch and yaw control, making it easier to design [2], [6].

Furthermore, in the proposed configuration, which is the twin-boom configuration, there is enough clearance from the propeller slipstream, thus ensuring minimal aerodynamic interference. The fact that the horizontal stabilizer is located between the booms ensures minimal interference from the wake behind the fuselage and wing. The twin-tail configuration is advantageous in that it has less aerodynamic interference at high angles of attack.

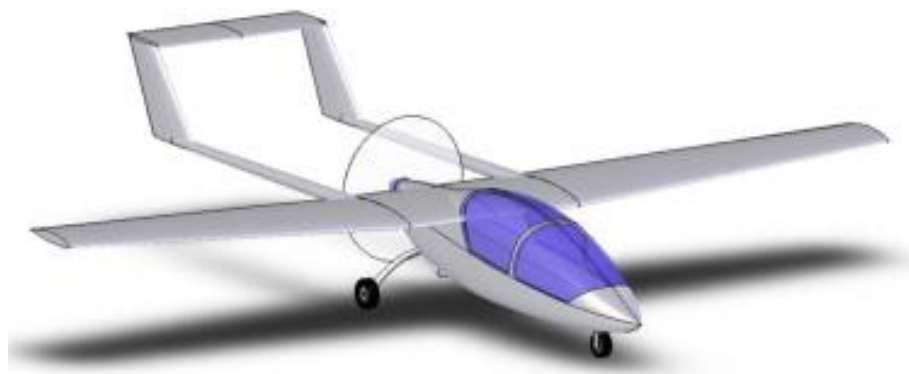


Figure 5.2 : Tail Selection [Courtesy: ScienceDirect]

5.3.Fuselage Configuration

The choice of a central pod fuselage configuration has been made for the THAR UAV in order to effectively house the payload, avionics, propulsion interfaces, and energy systems in an efficient manner, providing structural simplicity and aerodynamic cleanliness. The fuselage is designed as a compact primary body located underneath the high wing, which houses the essential components such as the EOIR turret, neutraliser payload system, flight control computer, communication systems, and energy storage units. The centralized fuselage ensures the effective distribution of the mass distribution and the center of gravity (CG), which is

essential for stable flight as well as the smooth transition between flight modes [2], [6]. The central pod fuselage configuration is best suited for the UAV as it houses the twin-boom tail arrangement effectively by allowing the tail loads to be carried by the booms rather than the fuselage, which is long in the UAV. This provides the UAV with a lighter and structurally efficient design as well as modularity. The fuselage could be designed in detachable sections as required by the UAV design, which is a significant consideration in UAV design [2]. Aerodynamically, the fuselage is made to be streamlined in order to reduce parasitic drag during the cruise phase, which is essential for the UAV's endurance performance. The underside of the fuselage is made to fit the belly-mounted payload systems in order to provide the UAV with an unobstructed field of view. Moreover, the positioning of the wing in relation to the fuselage ensures the UAV has sufficient ground clearance for improved safety [6]. The chosen fuselage design, which is the central pod fuselage, provides the UAV with the required balance in terms of payload integration, structural efficiency, aerodynamic performance, and operational practicality in order to achieve the required long-endurance UAV performance.

5.4.Cruise Propulsion System – Pusher Configuration

A rear-mounted pusher propeller configuration has been chosen for the THAR UAV in order to ensure efficient propulsion during forward flight while maintaining aerodynamic cleanliness of the forward fuselage. In the chosen configuration, the location of the propeller behind the wing and the fuselage ensures that the airflow over the wing and the payload area remains relatively undisturbed. This would be highly advantageous in surveillance missions, where it would ensure the UAV has an unobstructed field of view, in addition to minimizing aerodynamic interference in the forward area. From an aerodynamic point of view, the pusher configuration enables the wing to perform in a cleaner airflow compared to tractor configurations, thus enhancing the performance in terms of lift characteristics, among other attributes, making the UAV more aerodynamically efficient during cruise flight. This directly contributes to the accomplishment of long endurance performance, as efficiency is highly dependent on the maintenance of a high lift-to-drag ratio. Moreover, the positioning of the propulsion system at the rear integrates efficiently with the twin boom configuration, enabling the propeller to be mounted between the booms without interfering with the empennage, which is the standard configuration for UAV [5].

During the cruise flight phase, the propulsion system runs at optimized power settings, which are equivalent to the maximum endurance conditions. On the other hand, during the climb phase, the propulsion system uses higher power settings to produce the required excess thrust.

In addition, the absence of disturbances in the forward fuselage region due to the propeller improves sensor performance and minimizes forward noise propagation in the UAV, which is desirable in surveillance operations. The pusher propulsion configuration strikes a balance between aerodynamic efficiency, payload integration, and mission effectiveness in the UAV, making it desirable for long-endurance UAV operations [5].

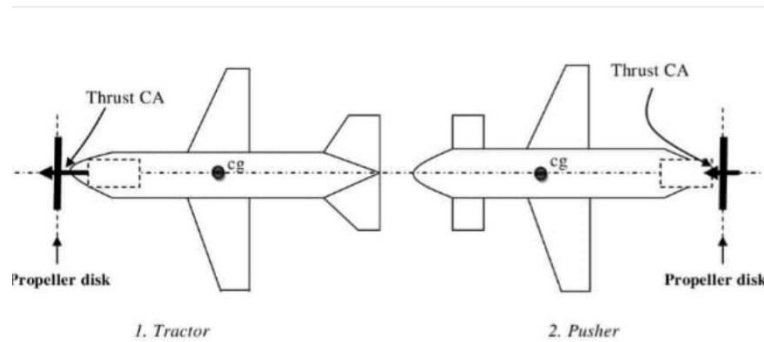


Figure 5.3 : Cruise Propulsion System [Courtesy: Aeromodelling Club IIT Kanpur]

5.5.Lift Propulsion System – Electrical VTOL Configuration

The VTOL mode of the THAR UAV is facilitated by the distributed electric lift propulsion system, which consists of four rotors that are mounted symmetrically around the UAV's center of gravity. This has been selected as it provides a reasonable compromise between system simplicity, control authority, and system redundancy. The rotors are arranged such that the resultant thrust vector acts close to the center of gravity, thereby minimizing pitching and rolling moments. The symmetric rotors facilitate differential thrust control, thereby achieving accurate attitude stabilization [2].

The distributed electric propulsion system has been designed to provide total thrust in excess of the UAV weight, thereby achieving stable hover and vertical climb conditions. Due to the high-power demand in hover, the system has been sized accordingly; however, the operational strategy minimizes the time spent in hover mode, thereby maximizing endurance. The purpose of the lift propulsion system is to produce total thrust greater than UAV weight so that there is sufficient thrust for stable hover and vertical climb. Due to the high-power demand in hover state, the system is designed to produce this power. In addition, the strategy ensures that minimum time is spent in hover state to conserve energy for long endurance.

During its operation, the lift propulsion system is used for vertical take-off until clearance is obtained. This ensures clearance of any obstacle and safe take-off. A transition phase is used where both the lift and cruise propulsion systems are used. This ensures a smooth transition

from rotor-borne lift to wing-borne lift. During this phase, initial climb is assisted by the lift propulsion system. However, the system is switched off when cruise is attained.

In the descent and recovery phase, the lift motors are utilized to execute a controlled vertical landing, which affords precise landing capability. This eliminates the need for runway infrastructure, making the UAV suitable for operation in unprepared or confined areas. From an aerodynamic perspective, the design combines the lift rotors to avoid interference during forward flight, with provisions to reduce drag when inactive. The electrical VTOL configuration affords a , flexible, and mission-adaptable solution while maintaining cruise efficiency.

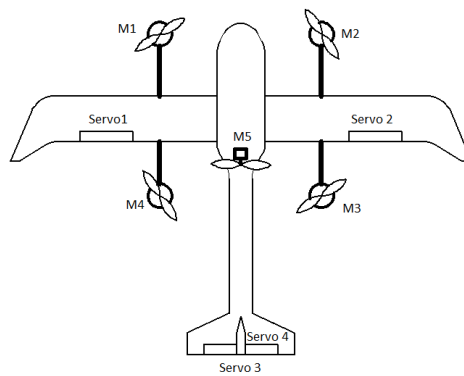


Figure 5.4 : Electric VTOL Propulsion System [7]

5.6. Payload Integration

The payload system of the THAR UAV includes an EOIR turret located at the forward section of the fuselage and a neutraliser capsule launcher located downstream on the fuselage's underside, a standard practice for UAVs. The EOIR camera is required to offer a 360-degree azimuth coverage with a high elevation range, thus requiring an unobstructed field of view in all directions, a principle of EO/IR payload design. During initial layout consideration, it was noted that when the neutraliser payload is placed at the same vertical level, it may cause a problem of field of view obstruction, especially when rotating rearward on the EOIR turret. Therefore, to overcome this problem, a vertical offset is provided below the fuselage's reference line for mounting the EOIR camera, thus keeping its rotational envelope unobstructed. This deliberate vertical separation of the payload systems ensures that there is continuous and unimpeded coverage, thus preserving surveillance effectiveness at all times. At the same time, this optimizes the position of the neutralizer system for safe and effective deployment without interfering with the sensors. Moreover, this forward placement of the camera optimizes the detection capability, and this separation of systems optimizes efficiency in operation. Proper

placement of the payload is very important since it affects the center of gravity, stability, and flight performance. In general, it is evident that a chosen UAV payload integration strategy strikes a delicate balance between performance, structural integrity, and efficiency in operation, thus assuring successful mission execution.

5.7.Landing Gear Configuration

For landing support, a simple landing support system is chosen for the THAR UAV, which is used for ground support and does not require a landing gear system. Since it is a VTOL flight, there is no requirement for wheels, taxiing, or large horizontal loads. Therefore, a lightweight and simple structure is chosen, keeping in mind that it is used for lifting the UAV above the ground. The chosen configuration is a fixed landing support system, known as a skid-type landing support system, without wheels. This is chosen for lifting the UAV above the ground, keeping in mind that a certain distance is required between the ground and the UAVs payload, propulsion system, and underside of the fuselage. A skid-type landing support system is commonly used for UAVs, and this is chosen for its simple design, allowing it to be used on uneven terrain.

The main purpose of this structure is to provide sufficient clearance on the ground and protect the sensitive components during take-off and landing. In addition, this structure provides the UAV with basic stability when it is on the ground, which ensures it does not tip over in pre-flight and post-flight positions. Since the UAV descends vertically using propulsion, the landing structure doesn't need to have shock absorbers. This simple landing structure using a skid-based mechanism is suitable for VTOL UAVs, as it provides the UAV with several benefits in terms of simplicity, weight reduction, and the ability to land in rough terrain without the need for a runway.

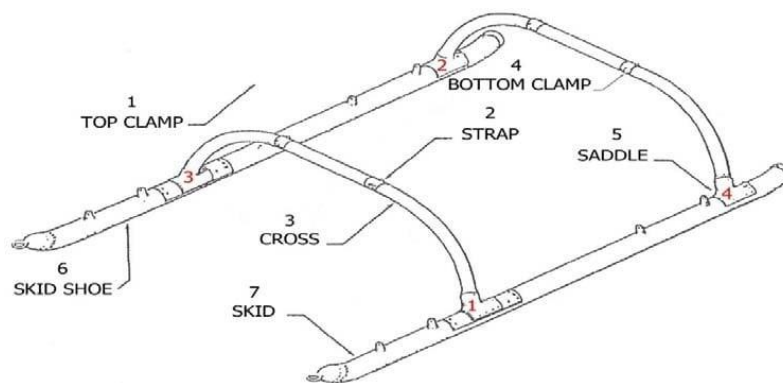


Figure 5.5 : Landing gear [Courtesy: Research Gate]

Chapter 6 Aerodynamic Characteristics

6.1. Aerodynamic Analysis

In conceptual UAV design, aerodynamic performance is mainly dependent on parameters such as lift, drag, and their combination with flight conditions. These parameters are mainly dependent on factors such as wing geometry, airfoil, and overall UAV configuration, etc. Main aerodynamic parameters: Main parameters that are used in this analysis are lift coefficient C_L , drag coefficient C_D , and lift-drag ratio L/D . These parameters are of significant importance in determining flight performance parameters such as endurance, efficiency, and power requirement, etc. [2], [5].

In the case of long endurance UAVs with moderate flight velocities, the aerodynamic efficiency represents a key design goal. This involves striking a balance between lift generation and drag minimization to ensure efficient flight over long periods of time. The aerodynamic estimation process developed in this section represents the basis for the performance analysis and design sizing of the UAV. The analysis process assumes standard atmospheric conditions in line with the requirements defined for the THAR UAV [6].

6.1.1. Wing Airfoil Aerodynamics

The aerodynamic characteristics of various airfoil options are analysed using XFLR5 to find the best airfoil for the THAR UAV. The analysis is performed at a Reynolds number of approximately 2×10^6 , which corresponds to the cruise flight condition of the UAV. Batch analysis is carried out for various airfoils such as NACA 23012, NACA 23015, NACA 2312, NACA 4412, and NACA 2412, and their performance is compared based on various parameters such as lift coefficient, drag coefficient, and lift-drag ratios [2]. The selection criteria include the ability of the airfoil to achieve the desired lift coefficient at cruise, aerodynamic efficiency, low drag, and smooth stall behaviour. From this comparison, it can be found that the airfoil NACA 23012 is selected as the best suitable airfoil because of its better performance in terms of generating lift and aerodynamic efficiency within the range of Reynolds number [5], [6].

The lift characteristics show that at an angle of attack between 3° to 5° , the lift coefficient required for cruising is obtained at a value of approximately $c_l \approx 0.3 - 0.5$. This is ensured that the UAV can be flown at steady level flight at moderate values of angle of attack. The lift coefficient vs. angle of attack for NACA 23012 airfoils is shown in Figure 6.1. From the drag characteristics shown in Figure 6.2, it is evident that the airfoil selected has a relatively low

value of drag at lower and moderate values of angle of attack. This is a definite advantage in enhancing UAV endurance and power requirements [2], [5].

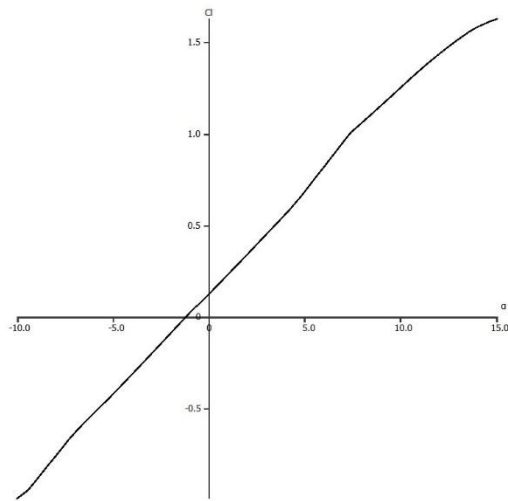


Figure 6.1 : c_l vs α (Airfoil)

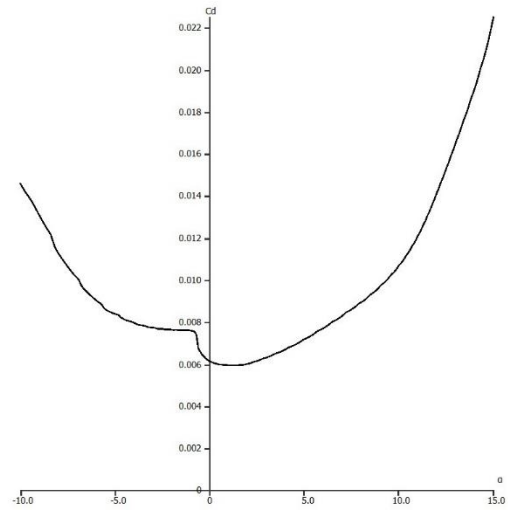


Figure 6.2 : c_d vs α (Airfoil)

The efficiency of the aerodynamics is maximum at the point where the angle of attack is between 8° and 10° , which is the cruise flight condition. The variation of the lift-drag ratio for the NACA 23012 airfoil is as shown in Figure 6.3. The comparison of the drag polar for the chosen airfoil further confirms that the chosen airfoil has less drag at the same lift coefficient compared to the other airfoils under consideration, which signifies the efficiency of the chosen airfoil in the context of aerodynamics. The drag polar curve for the NACA 23012 airfoil is as shown in Figure 6.4 [2], [5].

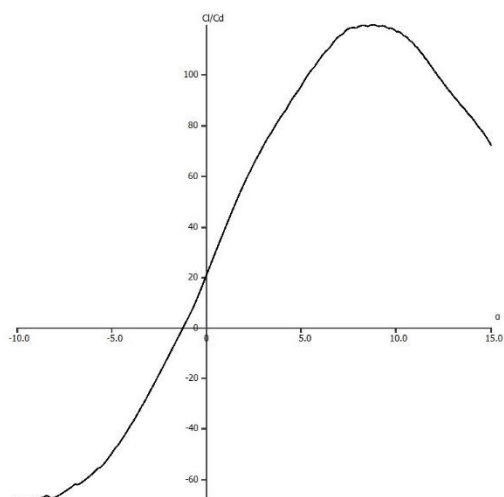


Figure 6.3 : c_l / c_d vs α (Airfoil)

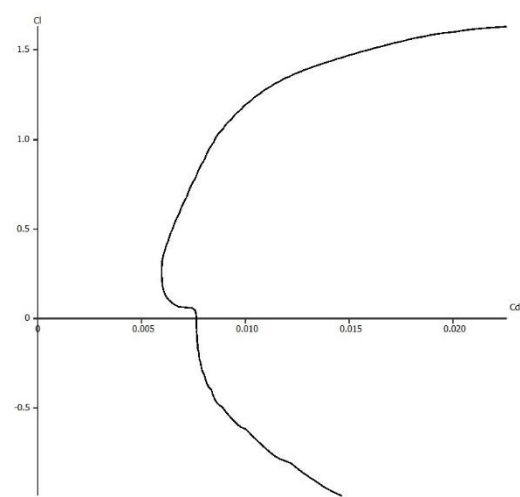


Figure 6.4 : c_l vs c_d (Airfoil)

The comparative performance of the selected airfoils with regard to aerodynamic characteristics is presented in Table 6.1. The table shows the Reynolds number, maximum lift coefficient, lift coefficient at cruise, drag coefficient at cruise, and the aerodynamic efficiency for each airfoil. On the basis of the comparison, the airfoil selected for the design is NACA 23012, as it can provide the required lift at cruise with better aerodynamic efficiency [2], [5].

Table 6.1 : Comparative Analysis of Airfoils

Airfoil	Reynolds number	c_l max	c_l at cruise	c_d at cruise	c_l / c_d max
NACA 23012	2×10^6	1.6	0.65	0.005	130
NACA 23015	2×10^6	1.7	0.70	0.0065	108
NACA 2312	2×10^6	1.5	0.60	0.0055	109
NACA 4412	2×10^6	1.7	0.80	0.0085	94
NACA 2412	2×10^6	1.6	0.7	0.0068	103

6.2.Wing Aerodynamics

The aerodynamic characteristics of the wing are determined by using XFLR5 to include three-dimensional effects like wingtip vortices and induced drag. Because of the tapered wing configuration, the Reynolds number is a function of the local chord length along the span of the wing; that is, it is higher at the root and decreases towards the wingtip. For this reason, different Reynolds number polars are used to ensure that the aerodynamic performance of the wing can be predicted at cruising conditions [5].

Unlike the airfoil, the finite wing has lower aerodynamic efficiency due to the induced drag. The lift curve plot demonstrates that at an angle of attack between approximately 3° to 5° , the required lift coefficient of about $C_l \approx 0.3 - 0.5$ is achieved for the cruise condition, which confirms the capability of the wing to sustain steady flight. The variation of the lift coefficient

with the angle of attack for the wing is shown in Figure 6.5. The drag characteristics show that the drag coefficient varies by increasing the angle of attack due to the combined effect of parasite and induced drag. The variation in the drag coefficient is prominent at higher lift coefficients because of the effect of the wingtip vortices. The variation in the drag coefficient with the angle of attack for the wing is shown in Figure 6.6. The choice of the aspect ratio as approximately 6 helps in reducing the induced drag, which improves the aerodynamic efficiency of the wing [2], [5].

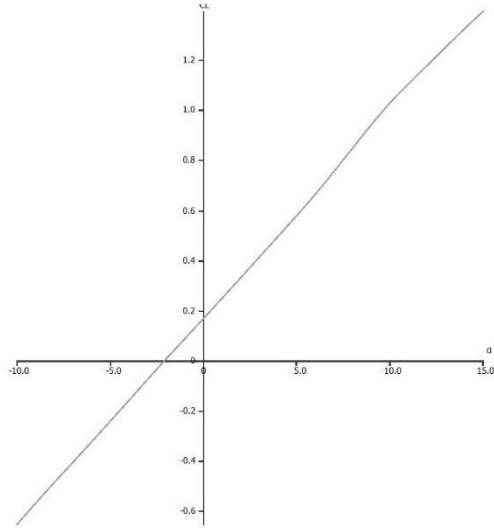


Figure 6.5 : C_L vs α (Wing)

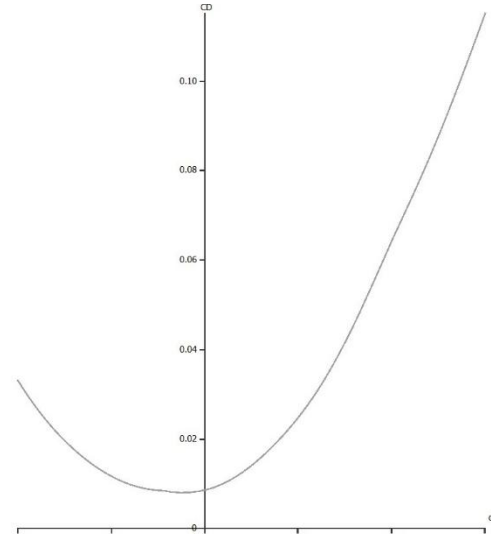


Figure 6.6 : C_D vs α (Wing)

The aerodynamic efficiency of the wing, given by $C_L/C_D \approx 25 - 28$, is maximum at an angle of attack of about 2° to 4° , which is the optimal cruise flight of the UAV. This implies that the wing is more efficient over this range, where a compromise is made in terms of lift creation and drag minimization. The variation of lift-drag ratio with respect to the angle of attack for the wing is depicted in Figure 6.7. The drag polar of the wing describes the relationship between the lift and drag characteristics. From this drag polar, it is evident that higher drag values are associated with the wing, compared to those of the airfoil, due to three-dimensional effects. However, this relationship is similar in both cases, and it is used to compute the required parameters, such as power requirement and endurances. The drag polar of the wing, given by C_D vs. C_L , is shown in Figure 6.8. [2], [5].

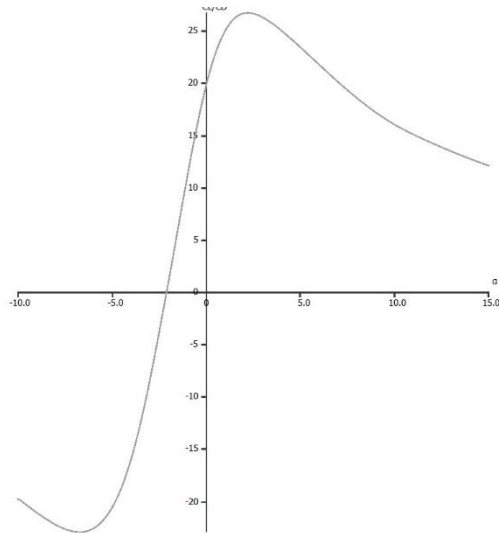


Figure 6.7 : C_l/C_d vs α (Wing)

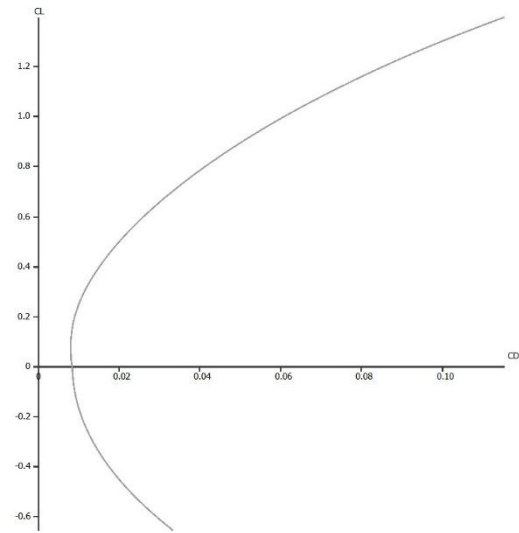


Figure 6.8 : C_l vs C_d (Wing)

The aerodynamic characteristics obtained from the analysis show that the performance of the selected wing configuration is satisfactory for the given flight conditions. From the lift curve, it is evident that the required lift coefficient is obtained at a moderate range of angle of attack. Thus, stable flight operation is ensured. In addition, the drag characteristics show a gradual increase in drag with an increase in angle of attack. This is due to the combined effect of parasitic and induced drag. Moreover, the lift-drag ratio characteristics show a distinct peak at a certain angle of attack. This shows that the selected wing configuration is satisfactory for the mission requirements since it is capable of operating at a condition of maximum aerodynamic efficiency. In this case, power consumption is minimized. Therefore, it is evident that the selected wing configuration is satisfactory for the mission requirements.

Chapter 7 Initial Sizing and Constraint Analysis

7.1. Mission Requirements and Design Inputs

The design requirements for the THAR UAV are directly taken from the problem statement of NACDeC-IX. The UAV is required to have a payload of 15 Kg, a cruising altitude of 3000 m, and a mission endurance of more than 9 hours. Furthermore, the UAV is required to take off from a 50 m x 50 m area without a runway and meet geometric constraints that include a maximum wingspan of 6 m and a maximum weight of 175 Kg or less.

7.2. Mission Profile

The mission profile of the THAR UAV is shown in Figure 7.1. It is evident that a series of flight phases are involved in this mission profile, which include VTOL take-off, climb, cruise, loiter, descent, and vertical landing. It is to be understood that this profile represents the mission envelope of the UAV.

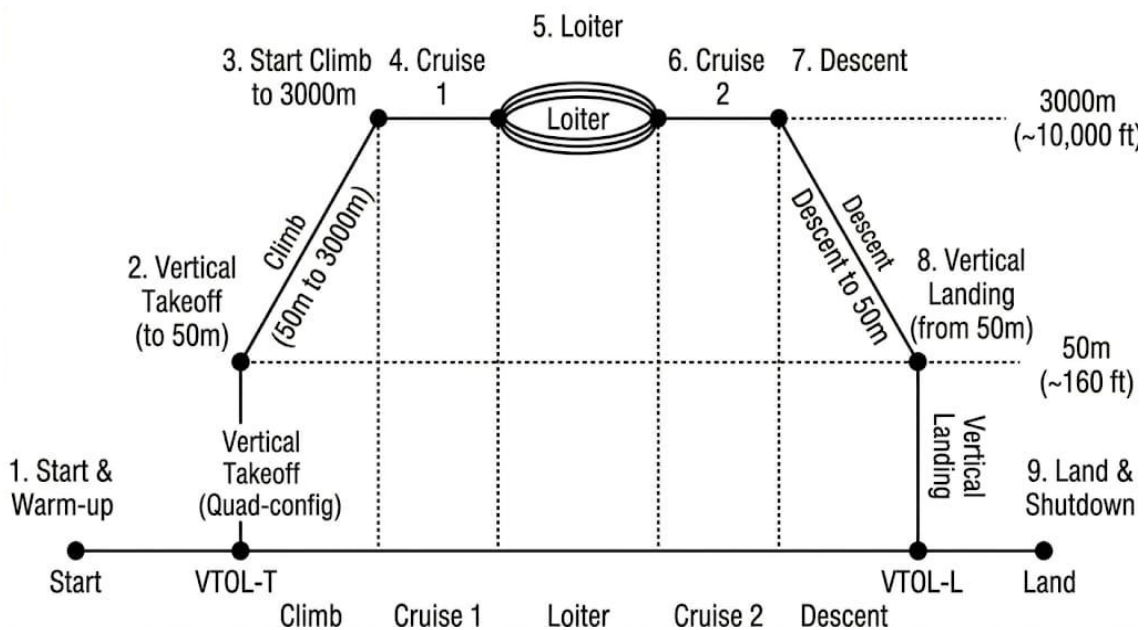


Figure 7.1 : Mission Profile

7.2.1. Start and Warm-up

This phase involves the initialization of the system, where all the subsystems on board the UAV, including the avionics, the communication system, the propulsion system, and the payload, are initialized. Diagnostics are also performed to ensure the reliability of the system, especially for long-endurance flights. Proper warm-up of the propulsion system ensures stability in the system, thus preventing failures during crucial stages of the flight.

7.2.2. Vertical Take-off

The UAV uses a vertical take-off mode using distributed electric propulsion, which enables the UAV to take off from a confined area without the need for a runway. The UAV takes off vertically into the air to a height of 50m, which ensures the UAV is clear of the ground obstacles, especially in desert environments, avoiding the ingestion of dust into the UAV. This phase requires high thrust and power, making it the most energetic phase in the mission.

7.2.3. Climb to Cruise Altitude

After the UAV has attained a safe altitude for transition, it changes mode of flight from vertical flight to forward flight with gradual transfer of lift force from the rotor thrust to aerodynamic lift. This is done in a controlled manner in order to maintain stability and power efficiency. The UAV then ascends from 50 m to 3000 m at an optimal rate. Flying at higher altitude ensures reduced drag, increased endurance, and improved surveillance with efficient flight conditions.

7.2.4. Cruise Phase – 1

In this phase of flight, the UAV cruises steadily and levelly at an altitude of 3000 m while moving towards the assigned surveillance region. In this phase of flight, the UAV cruises at an optimal speed that is equal to the maximum lift-drag ratio. This is the optimal speed for the UAV because it ensures the minimum fuel is consumed while in flight. In this phase of flight, the conditions are stable so that the structural conditions of the UAV are met and the propulsion conditions are met.

7.2.5. Loiter Phase

The loiter phase signifies the UAV's primary mission phase, in which the UAV flies in a controlled flight pattern over the target area. In this phase, the UAV flies at a constant altitude in circular patterns, also known as racetrack patterns, to cover the maximum area. In the loiter phase, the following operations are performed:

- Continuous surveillance using EO/IR systems
- Detection and verification of targets
- Real-time data transmission to the ground station

This phase requires the UAV to have high aerodynamic efficiency.

7.2.6. Cruise Phase – 2

After the completion of surveillance and payload deployment, the UAV leaves the loiter region and starts its return cruise to the deployment location. At this time, it is likely that there will be an improvement in aerodynamic performance and fuel efficiency as the weight of the payload is reduced. The UAV is still flying in an optimal cruise mode. The speed and height of the UAV remain constant. The energy is being consumed in an efficient manner.

7.2.7. Descent

The UAV undertakes a controlled descent from 3000 m altitude down to an altitude of around 50 m. This is done in accordance with a well-planned flight path. This phase of the flight ensures aerodynamic stability and does not allow the UAV to experience any high sink rate that would interfere with the control of the UAV. In this phase of the flight, the UAV operates on low power settings and makes use of gravity-assisted descent in order to save energy.

7.2.8. Vertical Landing

At low altitude, the UAV returns to the vertical flight mode. The lift propulsion system is again engaged, facilitating the descent in a precise and controlled manner. The vertical landing mode ensures safe landing in constrained areas, eliminating the need for prepared landing sites. Careful thrust control is required to ensure stability in this mode, avoiding hard landing.

7.2.9. Landing and Shutdown

Once the UAV has touched down, the propulsion systems are gradually shut down, and post-flight activities are performed. The data gathered during flight is secured, and transmission of data, if necessary, is done. The UAV is then made ready for disassembling and transportation, thus fitting well within the operational demands of deployment in a remote location.

7.3. Initial Weight Estimation

The initial weight estimation is carried out using the Class-I weight estimation method, which is widely applied in the conceptual UAV design phase for the estimation of the UAV weight in the first instance using statistical weight fractions. In accordance with the UAV weight estimation standards, the total weight of the UAV can be represented as:

$$W_0 = W_{payload} + W_{empty} + W_{fuel} \quad (7.1)$$

where W_0 is the Maximum Take-Off Weight (MTOW), $W_{payload}$ is the payload weight, W_{empty} is the empty weight, and W_{fuel} is the fuel weight. Based on similar Medium UAV

configurations, the empty weight fraction typically lies in the range of 0.45 to 0.55 of the MTOW [2], [6].

7.3.1. Payload Breakdown

The payload of this THAR UAV includes an EO/IR surveillance system and a neutraliser capsules deployment system, as described in the mission requirement. The total weight of the payload is fixed at 15 Kg, affecting the overall weight, CG location, and design of the UAV.

Table 7.1 : Payload Break Down

Component	Weight (Kg)
EOIR Turret	5
Neutraliser Capsules (4 × 1)	4
Launcher	6
Total Payload	15

7.3.2. Endurance-Based Fuel Fraction Estimation

For propeller-driven UAV, endurance is estimated using the Breguet endurance equation:

$$E = \frac{\eta_p}{c} \left(\frac{L}{D} \right) \ln \left(\frac{W_i}{W_f} \right) \quad (7.2)$$

where E is endurance, η_p is propeller efficiency, c is specific fuel consumption, L/D is lift-to-drag ratio, W_i and W_f are the initial and final weights respectively.

In order to perform the calculations, the lift-to-drag ratio is considered to be 20, the efficiency of the propeller is considered to be 0.8, and the specific fuel consumption is considered to be around $8.17 \times 10^{-8} \text{ s}^{-1}$. The fuel fraction is considered to be 0.267, which corresponds to a fuel weight of around 40 Kg. Using these parameters, it is found that the endurance of the UAV is around 16 hours which is 57,600 seconds, which is more than the required endurance of 9 hours including propulsion system corrections in real-time with considerable losses it is approximated to be around 11 to 13 hours of endurance [5].

7.3.3. Weight Fractions Assumptions

The distribution of total weights in structural weight, propulsion weight, avionics weight, fuel weight, and payload weight is normally estimated using statistical weight fraction data obtained

from similar UAV configurations. It is a reasonable approximation of UAV weight distribution at the early design phase of a conceptual UAV design process [2].

Table 7.2 : Weight Fraction

Component	Weight Fraction (MTOW)
Propulsion	0.086
Payload	0.1
Avionics	0.166
Fuel	0.266
Structure	0.38

The above values are selected based on typical medium UAV configurations and are used as an initial estimate for weight fraction distribution. These values will be refined in later stages of design.

7.3.4. Initial MTOW Estimation

As the actual weight of the UAV cannot be determined at the conceptual design phase, it has been assumed that the MTOW will be 150 Kg, which will be refined further at the subsequent stages of the design process.

Table 7.3 : MTOW Estimation

Component	Weight (Kg)
Propulsion	13
Payload	15
Avionics	25
Fuel	40
Structure	57
Total	150

7.4. Flight Speeds

The choice of flight speeds is important in the conceptual design of the UAV since it affects the aerodynamic performance of the UAV directly. The important flight speeds considered in the design are the stall speed and the cruise speed of the UAV. These are considered to be the limits of the UAV operation. These speeds are considered to be the primary inputs for aerodynamic performance estimation [5].

7.4.1. Stall Speed

The stall speed of an UAV is defined as the minimum speed at which an UAV can generate sufficient lift to balance its weight. Stall speed is a critical UAV design parameter for the purpose of UAV take-off, landing, and low-speed operations [2].

The stall speed is determined using the lift equation under maximum lift conditions:

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L_{max}}}} \quad (7.3)$$

where W is the UAV weight, ρ is air density, S is wing area, and $C_{L_{max}}$ is the maximum lift coefficient.

For the current UAV design, the stall speed is found to be 20.65 m/s. It is assumed that the UAV design requirements are satisfied at this speed. It is a general assumption that the stall speed is around 20.65 m/s during the design of a medium UAV [2], [5].

7.4.2. Cruise Speed

The definition of the cruise speed is the UAV's efficient flight speed during steady flight. At the cruise flight, the lift produced by the UAV's wing balances the UAV's weight. For a steady level flight, the Lift of an UAV can be expressed as:

$$L = W = \frac{1}{2} \rho V^2 S C_L \quad (7.4)$$

Therefore, the cruise speed can be obtained as:

$$V = \sqrt{\frac{2W}{\rho S C_L}} \quad (7.5)$$

where W is the UAV weight, ρ is air density, S is wing area, and C_L is the lift coefficient at cruise condition.

Using the above relation, the cruise speed is computed to be around 40 m/s. This cruise speed is necessary for achieving aerodynamic efficiency through a high lift-to-drag ratio while at the same time ensuring reasonable power demands for flight efficiency. The cruise speed is within the normal characteristics of small UAVs as discussed in references [2], [5].

7.5. Wing Loading Selection

Wing loading is one of the most significant parameters in UAV design, as it affects stall speed. Lowering the wing loading will lower the stall speed, which is necessary for a runway-independent flight within a specified area of 50 m. However, lower wing loading will increase drag, thus lowering efficiency in flight. Therefore, a suitable value of wing loading must be chosen [2].

In order to ensure the safe operation of the UAV in the limited take-off area, the wing loading has to be limited by the stall speed, which can be determined by the following equation:

$$\frac{W}{S} = \frac{1}{2} \rho V_{stall}^2 C_{Lmax} \quad (7.6)$$

where ρ is air density, V_{stall} is stall speed, and C_{Lmax} is the maximum lift coefficient.

In the preliminary design, the values of the variables are assumed as follows:

$V_{stall} = 20.65$ m/s, $C_{Lmax} = 1.3$, and $\rho = 0.909$ Kg/m³ (at 3000 m altitude). According to the above analysis, the wing loading of 251.951 N/m² has been determined as a compromise between the stall speed and the cruise efficiency [5].

7.6. Wing Geometry Determination

Once the wing loading is chosen, the parameters of the wing's geometry, such as the area, wingspan, and the mean aerodynamic chord, are computed in order to satisfy the aerodynamic and geometric constraints [5].

The wingspan is determined using the aspect ratio relation:

$$b = \sqrt{AR \cdot S} \quad (7.7)$$

where AR is the aspect ratio.

The mean aerodynamic chord (MAC) is calculated using:

$$\bar{c} = \frac{2}{3} C_r \frac{(1 + \lambda + \lambda^2)}{(1 + \lambda)} \quad (7.8)$$

where C_r is the root chord and λ is the taper ratio.

To satisfy the design constraint on the wingspan being less than 6m as required by the mission requirement specification, the aspect ratio is chosen in such a way that the wingspan is within the allowable limit. An appropriate aspect ratio is chosen as a compromise between the conflicting requirements of minimum induced drag and the feasibility of the design. The chosen wing geometry satisfies the mission requirement specification while providing the required aerodynamic efficiency.

7.6.1. Wing Parameters

The parameters of the wing are chosen in a manner that ensures aerodynamic efficiency, keeping in mind all the constraints. A moderate aspect ratio is chosen to ensure a high lift-drag ratio, which is a prerequisite for long-endurance flights. The geometry of the wing is chosen in a manner that ensures sufficient lift is generated during cruise flight. For this purpose, a tapered wing is chosen, keeping in mind that it will minimize induced drag and ensure a more efficient lift distribution. At the same time, it is kept simple. For this purpose, a zero sweep is chosen, keeping in mind that it will ensure good low-speed performance and will be easy to manufacture.

Table 7.4 : Wing Parameters

Parameter	Value
Span (m)	5.99
Wing area (m^2)	5.84025
Aspect ratio	6.1435
Root chord (m)	1.360
Tip chord (m)	0.590
Taper ratio	0.433
Mean Aerodynamic Chord (m)	1.0259
Wash In ($^\circ$)	3
Wash Out ($^\circ$)	-2

Chapter 8 Propulsion System

8.1. Selection of Propulsion System

The selection of an appropriate propulsion system for a Tactical Hunting and Reconnaissance (THAR) UAV depends on a variety of requirements and parameters associated with the successful accomplishment of the mission. Some of these requirements are the endurance and range of the UAV for surveillance, the speed at which the UAV cruises, the altitude at which the UAV operates, the independence from runways due to VTOL capability, the payload capacity associated with the UAV, the fuel efficiency associated with the operation of the UAV for long periods, the thrust to weight ratio associated with the propulsion system to enable VTOL capability, the hovering capability associated with the propulsion system to enable the precise engagement of the target, the aerodynamic efficiency associated with the propulsion system to ensure efficient operation, the capability to vary the altitude to enable the precise engagement of the target, the capability to efficiently operate in harsh environments, the capability to efficiently operate for a long period with minimum maintenance requirements.

The propulsion system for the THAR UAV is selected as a hybrid system, consisting of a VTOL system with four motors for vertical take-off and landing, and a fuel-powered pusher propeller system for cruise flight, in order to meet the runway-independent operation requirements. The selected configuration allows for the decoupling of vertical and horizontal propulsion, thus enabling independent optimization for each mode of flight and hence improving aerodynamic efficiency. In vertical take-off and landing, the propulsion system has to be able to develop a thrust force equal to or greater than the weight of the UAV.

8.2. VTOL Thrust Requirements

For an MTOW of 150Kg, the UAV weight (W) is 1471.5 N. This weight represents the total weight to be balanced by the propulsion system. In VTOL flight, unlike in regular UAV, the entire weight has to be balanced by thrust alone, without any contribution from aerodynamic lift. To ensure stable hover and safe vertical take-off, a safety factor for hover is considered. The safety factor for hover is given by $k = 1.5$. This safety factor includes other requirements such as control forces, thrust sapping effects, and external factors such as wind gusts [7]. The total thrust required is given by

$$T_{\text{total}} = k \times W \quad (8.1)$$

Where k is hover safety factor

Using this relation, it is calculated that the total thrust required is approximately 2207.25 N. This safety margin makes sure that the UAV does not only have the ability to hover but also has a safety margin for vertical climbing and disturbances during take-off or landing. In the chosen multi-rotor design, the total thrust is equally distributed among the rotors. In a design with four rotors, the thrust required for each of the rotors is approximately 552 N.

This is done so that each of the motors has a feasible range of thrust while also maintaining redundancy in the design. Furthermore, distributing the thrust across all the motors also makes it much more stable and easier to maneuver, especially during VTOL operations. The chosen thrust value makes sure that the propulsion system has the ability to maintain stable hover, vertical climbing, and landing safely while also maintaining a safety margin for real-world operations [2].

8.3.Cruise Thrust Requirement

The cruise thrust requirement for the THAR UAV is determined by the steady level flight conditions in which the lift generated by the wing equals the weight of the UAV, and the thrust generated by the propulsion system equals the drag on the UAV. In the steady level flight conditions, the thrust equals the drag on the UAV.

$$L = W \quad (8.2)$$

$$T = D \quad (8.3)$$

From these conditions we get,

$$T = \frac{W}{\left(\frac{C_L}{C_D}\right)_{max}} \quad (8.4)$$

From the aerodynamic analysis, the maximum lift to drag ratio is estimated to be in the range of 25 to 28. This is expected to occur at an angle of 3° to 5° , which is the optimal cruise condition. Using the aerodynamic efficiency, the required thrust for the optimal flight is estimated to be approximately 58.86 N. This is the minimum required thrust in ideal conditions. However, in the real world, there is the presence of additional drag due to the presence of the fuselage, the propulsion system, the VTOL rotors, and the interference effects. To account for the presence of the real-world effects, a correction factor is applied. In the case of the conceptual design phase, it is assumed to be 2.5. This is to account for the parasitic and interference effects. The corrected drag is estimated to be approximately 147 N. Additionally, to account for various uncertainties in the estimation of drag and efficiency of the propulsion

system, a design margin of 1.3 is used. This leads to a required thrust of approximately 191 N. Finally, to account for various other uncertainties and to ensure the performance of the system under all conditions, a design thrust of 200 N is selected for cruise conditions. This selected value of thrust is such that all the drag forces can be overcome while ensuring long endurance operation of the vehicle [7].

8.4.Cruise Power Requirement

The required cruise power for the THAR UAV is computed based on the thrust required to compensate for the drag force during steady flight. During cruise flight, the propulsion system should provide enough power to produce the required thrust at the flight velocity. The required power is given by:

$$P_{req} = T \times V \quad (8.5)$$

Where T is Cruise thrust and V is Flight velocity

Using the above-mentioned relationship and considering the cruise thrust to be 200 N and cruise velocity to be 40 m/s, the required power is calculated to be around 8 kW. This is the ideal power required for the UAV to operate at steady state. However, due to various losses such as inefficiency of the propellers and installation effects, the actual power required for the UAV to operate will be different from the ideal power calculated above. In order to account for these losses, the efficiency of the propellers is taken into consideration. The efficiency of the propellers is usually between 0.75 and 0.85 for small UAV propulsion systems. Hence, the actual power required by the UAV is higher than the ideal calculated power. After considering all these factors, a propulsion system is selected for the UAV such that the power rating of the propulsion system is around 8 kW. This ensures that the UAV is capable of steady cruise state while also considering various operating conditions such as changes in altitude, payload variations, and aerodynamic effects. Moreover, the UAV is also capable of operating at the optimal efficiency region of the propulsion system for long-endurance mission requirements [5], [7].

8.5.Propeller Aerodynamics

The aerodynamic performance of the propeller is significant in the determination of the thrust generation capability as well as the efficiency of the UAV. The propeller uses the rotational energy to generate thrust by accelerating the air flow, and the performance is based on the parameters such as advance ratio, blade geometry, etc. [2]. The performance of the propeller is

tested at the cruise condition with the velocity of approximately 40 m/s and the rotational speed of approximately RPM is 5000 – 6000. The aerodynamic performance is characterized by the thrust coefficient as well as the power coefficient, depending on the operating conditions. The efficiency of the propeller is given by propeller efficiency, defined by the ratio of output power to input power. From this analysis, it is estimated that the propeller efficiency is given by $\eta \approx 0.75\text{--}0.85$, indicating that the propeller is able to generate thrust in an efficient manner. The propeller efficiency characteristics were taken from Figure 8.1, which shows the variation of propeller efficiency with advance ratio for a three-bladed propeller with Clark Y sections at different blade pitch angles, and this graph was used for estimating the propeller performance parameters [5].

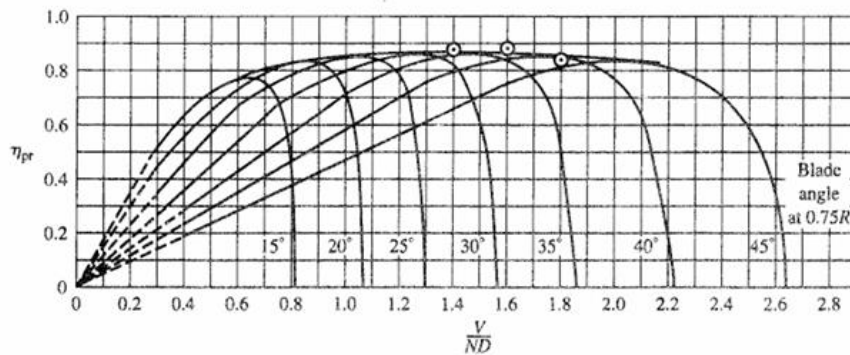


Figure 8.1 : Advance Ratio [5]

The following parameters were kept constant for the analysis. The advance ratio was taken as 0.9. Also, it was assumed that the required thrust is 200 N.

Table 8.1 : Propeller Efficiency vs Advance Ratio

Parameter	Value
Advance ratio	0.9
Thrust required (N)	200

Using the advance ratio defined above Table 8.1, the required propeller diameter is calculated using the relation:

$$J = \frac{V}{ND} \quad (8.6)$$

where V is the flight velocity, N is the rotational speed in revolutions per second, and D is the propeller diameter. The flight velocity is fixed at 40 m/s, while the rotational speed is chosen

based on the engine performance characteristics. From the performance data of the engine, a power output of 8 kW is considered with an RPM of approximately 4900. Under the above conditions of operation, the thrust and power coefficients are obtained from the propeller performance graph for the selected advance ratio [5].

The thrust and power generated by the propeller are calculated using the following non-dimensional equations:

$$T = C_t \rho n^2 D^4 \quad (8.7)$$

$$P = C_p \rho n^3 D^5 \quad (8.8)$$

where C_t is the thrust coefficient and C_p is the power coefficient obtained from the graph, ρ is the air density, n is the rotational speed in revolutions per second, and D is the propeller diameter. Using the values of C_t and C_p the propeller diameter is found in a manner that satisfies the conditions of required thrust and power. The calculations are done using MATLAB, and the parameters obtained are shown in Table 8.2.

Table 8.2 : Calculated Propeller Performance Parameters

Parameter	Value
Propellor Diameter (m)	0.544
Power coefficient	0.46
Thrust coefficient	0.409

Chapter 9 Basic Sizing of Control Surfaces

9.1.Introduction

The tail surfaces are of significant importance in giving stability and control to the UAV in terms of longitudinal and directional stability. The horizontal tail gives stability in pitch, whereas the vertical tail gives stability in direction. Proper sizing of the tail is of significant importance in achieving stability for flight, as well as control, over the entire mission range of flight [6].

9.2.Tail Volume Coefficient Method

The size of the tail surfaces can be determined through the use of the tail volume coefficient method, which relates the stabilizer area to the wing reference parameters. This design approach has been widely used in conceptual UAV design due to its simplicity in determining the UAV size during the initial stages of design [2].

9.2.1. Horizontal Stabilizer Sizing

The horizontal stabilizer is used for longitudinal stability and trim. The formula for determining the required horizontal tail area is given by:

$$S_H = \frac{V_H \cdot S_W \cdot \bar{c}}{l_H} \quad (9.1)$$

where S_H is the horizontal tail area, V_H is the horizontal tail volume coefficient, S_W is the wing area, $\bar{c}$ is the mean aerodynamic chord, and l_H is the horizontal tail moment arm.

For UAV configurations, the range of the horizontal tail volume coefficient should fall between 0.3 and 0.7. The V_H value of 0.4 is chosen to provide good longitudinal stability and trim ability, while keeping the UAV compact and efficient. [6]. In addition, sufficient robustness is obtained against any disturbances that might occur during long-endurance flights [2]. The length of the horizontal tail moment arm is selected such that it is a multiple of the mean aerodynamic chord length. This is given by:

$$l_H \approx 3\bar{c} \text{ to } 5\bar{c} \quad (9.2)$$

The longer the length of the horizontal tail moment arm, the better stability is obtained. In this case, a smaller tail area is obtained. However, this results in an increase in structural weight. Therefore, an optimal value is selected [6].

9.2.2. Vertical Stabilizer Sizing

The vertical stabilizer is used to achieve directional stability. This ensures stability in flight despite any disturbances that might occur, such as crosswinds and asymmetric loads. The formula used to calculate the required vertical tail area is given by:

$$S_V = \frac{V_V \cdot S_W \cdot b}{l_V} \quad (9.3)$$

where S_V is the vertical tail area, V_V is the vertical tail volume coefficient, S_W is the wing area, b is the wingspan, and l_V is the vertical tail moment arm.

For most UAVs, the range of the vertical tail volume coefficient will be 0.03 to 0.06. A value of V_V value of 0.035 will be used. This will ensure that directional stability is provided with an adequate balance of stability effectiveness and drag penalty [2]. The moment arm of the vertical tail will normally be assumed to be approximately equal to the moment arm of the horizontal tail to provide a compact UAV configuration with a high degree of structural efficiency.

9.2.3. Tail Parameters

The values of the tail parameters, which are decided on the basis of empirical design guidelines, are given in Table 9.1. These values are to ensure sufficient longitudinal and directional stability, along with aerodynamic efficiency, for the UAV. These values will be further refined in later stages of design [2].

Table 9.1 : Tail Parameters

Parameter	Horizontal Stabilizer	Vertical Stabilizer
Span (m)	2.6	0.55
Root chord (m)	0.35	0.5
Tip chord (m)	0.35	0.35
Taper ratio	1	0.7
Mean Aerodynamic Chord (m)	0.35	0.429
Area (m^2)	0.91	0.233
Aspect ratio	7.428	1.298

9.3. Control Surface Sizing

Control surfaces are provided to ensure proper maneuverability and stability in all three axes. The roll axis is controlled by ailerons, pitch by an elevator, and yaw by a rudder. These surfaces are designed as a fraction of the corresponding wing and tail surfaces to ensure proper maneuverability in all phases of flight [2]. The guidelines for designing control surfaces for conventional UAV and UAVs are based on various parameters such as surface area ratios, chord ratios, span ratios, and limits for deflection. These parameters influence the performance of each control surface.

9.3.1. Control Surface Design Parameters

The sizing of the control surfaces is carried out using empirical guidelines based on existing UAV and UAV configurations. The parameters include surface ratios, chord ratios, span ratios, and deflection limits. These factors define the effectiveness of the control surfaces [8].

Table 9.2 : Control Surfaces sizing parameters^[8]

Design Parameter	Aileron	Elevator	Rudder
Span Ratio	$\frac{b_A}{b} = 0.20 - 0.40$	$\frac{b_E}{b_H} = 0.80 - 1.0$	$\frac{b_R}{b_V} = 0.70 - 1.0$
Chord Ratio	$\frac{c_A}{c} = 0.15 - 0.30$	$\frac{c_E}{c_H} = 0.20 - 0.40$	$\frac{c_R}{c_V} = 0.15 - 0.40$
Surface Area Ratio	$\frac{S_A}{S} = 0.03 - 0.12$	$\frac{S_E}{S_H} = 0.15 - 0.40$	$\frac{S_R}{S_V} = 0.15 - 0.35$

9.3.2. Selected Control Surface Parameters

The initial dimensions for the control surfaces can be obtained using the control surface ratios selected, along with the previously calculated wing and tail geometry. The parameters obtained are presented in Table 9.3.

Table 9.3 : Control Surfaces Parameters

Parameter	Aileron	Elevator	Rudder
Span (m)	1	2.3	0.5
Chord Length (m)	0.2	0.15	0.15
Area (m^2)	0.2	0.345	0.075

Chapter 10 Conclusion

This Initial Design Report, which outlines the conceptual design process for the proposed THAR UAV, designed to satisfy the mission demands associated with persistent border surveillance within the Thar desert. The design process has been carried out with an emphasis on, with every design decision being directly related to the mission demands. The design process began with an understanding of the mission constraints, which influenced the selection of the overall configuration. The proposed configuration is a high-wing twin boom with an inverted U-tail, which is an appropriate solution considering the need for stability, payload, and propulsion. The proposed configuration will allow for ground clearance for the payload system, as well as the propulsion system, which is located at the rear.

For the aerodynamic configuration, the design was done using the NACA 23012 airfoil, which has good lift and stability characteristics. For the wing configuration, the choice was informed by the need to balance the conflicting requirements for aerodynamic efficiency and structural viability, especially in relation to the limited span. From the analysis, it is clear that the UAV is viable for the generation of the required lift while at the same time providing acceptable drag for cruise and loitering.

For the propulsion configuration, based on the calculation of Thrust required and Power required for the proposed UAV, so hybrid propulsion system has been chosen for the UAV. This is informed by the need to balance the requirements for vertical take-off and landing and cruise. For vertical take-off and landing, electric propulsion is chosen for its ability to operate in restricted spaces. For cruise, a pusher propeller has been chosen for its efficiency. This configuration has been chosen for its ability to balance the requirements for efficiency and viability. From the calculations, it is clear that the chosen configuration is viable for the mission.

Initial sizing and constraint analysis have been performed to determine parameters such as MTOW, wing loading, and power requirements. The results obtained are within the limits, indicating that the design will be able to perform the mission requirements. The mission profile, including climb, cruise, loiter, and return flight, has been taken into consideration to ensure proper flight of the UAV throughout the mission. Basic sizing of the tail and control surfaces has been performed to ensure proper stability and control of the UAV. The design selected will ensure proper behavior of the UAV, which is necessary in autonomous flight. The landing system has been designed as a simple support structure, as the UAV will be in VTOL mode.

Overall, the proposed design is a well-balanced solution that considers all the aspects of aerodynamics, propulsion, structure, and operational performance and stability. Rather than concentrating on the optimization of a specific parameter, the design is intended to ensure satisfactory performance in all aspects.

This report is the starting point in the design process. Further work in the design process will include more detailed analysis of weight estimation, stability and control, optimization of the propulsion system, and integration of the system. This will further improve the design and move it closer to a feasible and implementable UAV system.

In Conclusion, the THAR UAV design developed in this report meets the required design objectives and is a good starting point for further development in the subsequent stages of the design process.

References

- [1] R. Austin, *Unmanned Aircraft Systems: UAV Design, Development and Deployment*, 1st ed., Chichester: John Wiley & Sons Ltd, 2010.
- [2] D. Raymer, *Aircraft Design: A Conceptual Approach*, 2nd ed., Washington, DC: AIAA, 1992.
- [3] Israel Aerospace Industries, “Heron UAV,” Israel Aerospace Industries (IAI), 2023.
<https://www.iai.co.il/p/heron>.
- [4] JOUAV, “CW-80E UAV,” JOUAV, 2024.
<https://www.jouav.com/products/cw-80e.html>.
- [5] J. Anderson, *Aircraft Performance and Design*, 1st ed., New York, NY: McGraw-Hill, 1999.
- [6] M. Sadraey, *Aircraft Design: A Systems Engineering Approach*, 1st ed., Chichester, West Sussex, United Kingdom: John Wiley & Sons Ltd., 2013.
- [7] S. Gudmundsson, *General Aviation Aircraft Design: Applied Methods and Procedures*, 1st ed., Oxford: Butterworth-Heinemann (Elsevier), 2013.
- [8] UWARG, “Aileron Sizing,” University of Waterloo Aerial Robotics Group,
<https://uwarg-docs.atlassian.net/wiki/spaces/MEC/pages/2665349125/Aileron+Sizing>.
- [9] K. Rajendran, “Conceptual Design and Aerodynamic Analysis of Medium Altitude Long Endurance VTOL Fixed Wing UAV,” *Accelaron Aerospace Journal*, vol. 5, no. 4, p. 1477–1515, 2025.
- [10] J. K. Gunarathna and R. Munasinghe, “Development of a Quad-rotor Fixed-wing Hybrid Unmanned Aerial Vehicle,” 2018.